# GALOIS PROPERTY GOES UP IN ALGEBRAIC TOWERS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: The up-direction of the Galois-permanence package; the down-direction needs additional hypotheses and is treated separately in the chapter.

*Editorial note.* The label lemma-galois-goes-up names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Chapter lemma-normal-goes-up (source lemma): normality goes up in algebraic towers
- Chapter lemma-separable-goes-up (delivered card L48370): separability goes up in algebraic towers
- Chapter convention: Galois means normal and separable

*Source reference.* Original reference 033C: sources/stacks-fields/pinned-original-source.tex, source lines 2055–2055. Prerequisite pointer invoked at 2917; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09H2: sources/stacks-fields/pinned-original-source.tex, source lines 1404–1404. Prerequisite pointer invoked at 2917; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $K/E/F$ be a tower of algebraic field extensions. If $K$ is Galois over $F$, then $K$ is Galois over $E$.*

*Proof.* Combine Lemmas 033C and 09H2. □

*Date*: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.